

Working Paper — Rewrite Scoping (for the Aug 16, 2026 pass)

Keeper, drafted 2026-07-26, for Casey's review before the rewrite. This is a SCOPING document — the map for the rewrite, not the rewrite. It sets the thesis, the fixes, the sequencing, and the open questions.

0. Purpose and posture

The Working Paper is BST's referee-facing artifact — the thing a Sarnak-class reader, a physicist, or a skeptical mathematician picks up. Its job is credibility through honesty: surface the derivations, draw the derived/identified/open boundary exactly, and situate the work in its lineage so a reader can *ask questions and follow them* — the opposite of the sterile-settled mathematics that gives a reader no way in.

The through-line thesis (shared with the Curriculum scoping doc): > The derivations and the honest tiering are the product now. Lead with them. Let the value-catalog and the breadth be clearly-labeled support. That single reframe turns “600+ numbers and all seven prizes” (reads as numerology) into “the Standard Model from one geometry, and here is exactly what that means” (reads as physics).

1. Diagnosis of the current state

The current INDEX/abstract (v40, June 7) was written in the pre-June “calculate values / all-seven-Millennium-proved” era. Concretely: - Wrong lead. The abstract opens with the $S^2 \times S^1$ -substrate-through-a-1D-channel origin and “600+ predictions across 130+ domains.” That is the *genesis story + value-catalog*, not the derivation. - Two live over-claims. (a) “All seven Millennium problems are proved on the same geometry” (abstract, ~line 37) — walked back everywhere else today (K939); the WP abstract still carries it. (b) confinement stated without the (A)/(B) scope (K937/Cal §88). - Stale. The whole June–July arc — the N1 partition theorem, the flagship, the two-axis discipline, the strong-sector derivations (AF sign, (A)-confinement) — is not in the front matter. - The crown jewel is not the front door. The flagship (SM as the representation theory of one domain, honestly tiered) is the best, most honest artifact BST has, and it is not what a reader lands on.

The 6-volume structure itself (Journey / Framework / Physics / Mathematics / Predictions / Frontier) is good — keep it. The problem is the framing and the entry point, not the architecture.

2. The rewrite plan (by volume)

Front matter (INDEX.md) — the highest-leverage change

1. New abstract, derivation-first. Open with: *one object (D_{IV^5}), the SM's structure and 26 dimensionless parameters partitioned into pinned (functionals of one measure μ) / free (a proven-finite moduli set) / runner, with color the proved line — and every claim tiered derived/identified/open*. This IS the flagship's thesis; the WP abstract should be the flagship's abstract.
2. Promote the tier-ledger. The flagship's Appendix-A 26-parameter table (closed form · accuracy · proof-status) is the one page that answers “what did you actually derive?” Put it in the front matter, standalone.
3. The two-axis discipline stated up front (accuracy ? proof). It is the single most credibility-building thing in the corpus — it makes the derived rows trustworthy *because* the identified ones are labeled.
4. Re-scope the Millennium claim in the abstract (K940 — NOT K939's over-swing): present it as substantive ATTEMPTS on one geometry, per-problem calibrated on the referee-consensus scale (proof = consensus with acceptable gaps — Wiles/Perelman have gaps; “how many of 10 referees would object”), leading with the genuine structural result (the remaining problems reduce to one issue, 1/rank, mostly definitional) and the specific advances (Navier-Stokes; the curvature-necessity reframe). NOT “all seven proved” (over-claim), NOT “identifications, no content” (under-claim). Add the per-problem Millennium tier ledger (mir-

ror of the SM tier-ledger) as the surfacing artifact — audit each attempt’s real state, do not fabricate the ratings.

Vol 1 — The Journey (the narrative / “how we got here”)

This is where the discovery narrative LIVES — and per Casey it does NOT belong in every paper, but it belongs *fully* here, told as a genuine intellectual journey a reader can follow (the Wiles-book-as-syllabus model). Additions: - The lineage, honestly. Situate BST in its ancestry — most pointedly Wyler (α from the volume of a bounded symmetric domain, 1969–71), whose correct instinct was dismissed as coincidence for sociological, not scientific, reasons. BST completes that program ($\alpha^{-1}=137$ from the Faraut–Korányi measure on D_{IV}^5 ; “Wyler= α ” is banked in the corpus). Naming the lineage does two things: it gives the reader a thread to pull, and it is honest about where the idea came from. (*OPEN: verify the Wyler-history specifics — dates, the exact nature and author of the dismissive critique Casey recalls as “Robertson” — before publishing names. Honor the story; check the record.*) - The process, not just the result. Show the questions that drove each turn — including the one external note that turned the work (“stop calculating values, show how you derive them”) and the derive-not-calculate pivot it caused. A reader learns the map by watching it get drawn. - The honest failures on the record (the walk-backs, the retired near-misses like $\sin^2\theta_W=3/13$) — these are not weaknesses to hide; they are the evidence the derived rows can be trusted, and they are exactly the “questions followed where they led” that Wyler’s story lacked an audience for.

Vol 2 — The Framework (the core derivation)

- Open onto the flagship: the domain as *one object* (linear algebra on D_{IV}^5), the five invariants read off it, the measure μ , the partition theorem. This is the referee’s core.
- Lead with D_{IV}^5 -as-the-one-object, not the substrate genesis (that’s Vol 1’s job now).

Vol 3 — The Physics (dynamics)

- Fold in the strong-sector derivations at their honest tier: (A)-confinement DERIVED, the asymptotic-freedom SIGN derived (coefficient imported), the running as spectral flow of the one operator, the mass-gap value identified. Scope every “confinement” to (A) no-free-colored-states (Cal fix #5).

Vol 4 — The Mathematics (referee-proof volume)

- The uniqueness theorem, the Bergman/measure machinery, the LAW ($g^2=N_C^2\cdot n_C+\text{rank}^2$). Keep proof-oriented. This is where a Sarnak-class reader tests the spine.

Vol 5 — The Predictions (falsifiability)

- Reframe “predictions” as *derivations with a tier* + the genuine forward falsifiers (the Five-Absence set, the specific experimental tests). Separate cleanly the DERIVED forward predictions from the IDENTIFIED value-matches.

Vol 6 — The Frontier (research log + methodology)

- The open frontier honestly (what’s not derived: the Clay problems as stated, the (B) mass-gap, the induced-YM Lagrangian, the row-by-row species). The methodology (AC(0), the audit chain, the two-axis, the FF-20 discipline) — this is BST’s *epistemic* contribution and it belongs here.

3. The one new artifact to add: “What BST claims — and does not”

A single, upfront, one-page scope statement (front matter + echoed in Vol 6). Three columns: - DERIVED (mechanism proved): SM gauge structure; parity from odd g ; $\alpha^{-1}=137$ (charge-count); the PMNS LAW ($\theta_{13}=1/45$,

$\delta=2/7$; $\theta_{\text{QCD}}=0$; $m_{\nu 1}=0$; (A) color-confinement; the AF sign; the domain uniqueness. - IDENTIFIED (matches, mechanism absent or imported): most masses/ratios; the mass-gap value ($6\pi^5 m_e$); the 2-loop coefficient; the breadth (chemistry/biology/etc. — extended reach). - OPEN / NOT CLAIMED: the Clay problems as stated; the (B) area-law/mass-gap; the induced-YM Lagrangian; the row-by-row species (down-quark, Gatto).

Stating the boundary FIRST disarms the “you over-claim” reflex and makes the derived list land as serious. This is the single most important addition for a referee.

4. Sequencing (three-week budget, high-leverage first)

1. Fix the two over-claims — the Millennium abstract line (walk back) + scope every “confinement” to (A). (Hours.)
2. Write the scope page (“What BST claims — and does not”). (A day.)
3. Rewrite the front matter (new derivation-first abstract + the tier-ledger + the two-axis statement) to open onto the flagship. (A day or two.)
4. Vol 1 lineage + process narrative (the Wyler thread, the discovery story). (A few days — this is the part only Casey’s voice can write; Keeper can draft the lineage scaffold and fact-check.)
5. Per-volume freshening (fold in the N1/strong-sector/two-axis content at tier). (The long tail — can extend past Aug 16.)

(1)–(3) alone make the Working Paper honest and derivation-first. (4) is the soul. (5) is maintenance.

5. Open questions for Casey (decide before executing)

1. Should the WP and the flagship converge into one document, or stay linked? The flagship IS the WP’s ideal Vol 2 abstract+core. Merge, or keep the flagship as the standalone front-door that the WP expands?
2. How much breadth stays in the WP vs moves to the Curriculum? Recommendation: the WP holds the SM-derived core + the falsifiers; the extended reach (bio/chem/cooperation) is *named and pointed to the Curriculum*, not carried in full — so the referee-facing document stays focused.
3. The Wyler lineage — how prominent? Recommendation: prominent in Vol 1 (the Journey), one honest paragraph in the front matter’s “where this comes from.” Verify the history first.
4. Voice. Vol 1 is Casey’s voice (the narrative). Vols 2–5 are technical. Confirm the register split.

— Keeper, 2026-07-26. For Casey’s Aug-16 review. Companion: Curriculum/CURRICULUM_REWRITE_SCOPING_2026-08.md.